Metric Markov cotype two of ℓ_1 with constant 3024: a resolvent proof

Abstract

We prove that ℓ_1 has metric Markov cotype two with constant $C^2 = 3024$. For every stochastic matrix A that is reversible with respect to a probability vector π, every time horizon t and all points $x_1, \dots, x_n \in \ell_1$, there are points y_i such that $\sum_i \pi_i \|x_i - y_i\|_1^2 + t \sum_{i,j} \pi_i a_{ij} \|y_i - y_j\|_1^2$ is at most 3024 times the Cesàro-averaged energy of the x_i. The proof follows the resolvent approach of Mendel–Naor. It has three parts: a cut decomposition of finitely many ℓ_1 vectors, a "flat" cubic map whose squared increments are controlled by a quartic Hilbert potential, and a comparison of the geometric (resolvent) average with the Cesàro average, which costs a factor 28. Instead of a stopped martingale, the formal proof uses deterministic identities for the resolvent matrix $G = \sum_s (1-q) q^s A^s$. Every lemma and the main theorem are machine-verified.

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1 Setting

Throughout, $\mathbb{N}_0 = \{0, 1, 2, \dots\}$, $[n] = \{1, \dots, n\}$, and $\mathcal{P}_n$ is the set of all subsets of $[n]$, including $\emptyset$ and $[n]$. For $B \in \mathcal{P}_n$ and $i \in [n]$ we write $\mathbf{1}_B(i) \in \{0, 1\}$ for the indicator of $i \in B$. For an $n \times n$ real matrix A we set $A^0 = I$ and write $\delta_{ij} = I_{ij}$.

The space ℓ_1. We write ℓ_1 for the real Banach space of real sequences $x = (x(k))_{k\in\mathbb{N}_0}$ with $\|x\|_1 = \sum_{k\geq 0} |x(k)| < \infty$. Indexing by $\mathbb{N}_0$ rather than by $\{1, 2, \dots\}$ is immaterial. All norms of elements of ℓ_1 below are this norm.

Probability vectors and matrices. Let $n \in \mathbb{N}_0$.

- A vector $\pi \in \mathbb{R}^n$ is a *probability vector* if $\pi_i \geq 0$ for all i and $\sum_i \pi_i = 1$.
- A matrix $A \in \mathbb{R}^{n\times n}$ is *stochastic* if $a_{ij} \geq 0$ for all i, j and $\sum_j a_{ij} = 1$ for all i.
- A is *reversible with respect to* a vector $\pi \in \mathbb{R}^n$ if $\pi_i a_{ij} = \pi_j a_{ji}$ for all i, j.

These conditions are consistent: for $n = 1$, $\pi = (1)$ and $A = (1)$ satisfy all three.

Stationarity. Let A be stochastic and reversible with respect to an arbitrary vector $\pi \in \mathbb{R}^n$. Then for every $s \in \mathbb{N}_0$ and every j,

$$\sum_i \pi_i (A^s)_{ij} = \pi_j. \tag{1}$$

Indeed, for $s = 0$ this is trivial. If it holds for s, then

$$\sum_i \pi_i (A^{s+1})_{ij} = \sum_l \Big(\sum_i \pi_i (A^s)_{il}\Big) a_{lj} = \sum_l \pi_l a_{lj} = \sum_l \pi_j a_{jl} = \pi_j.$$

Reversibility enters the whole proof only through (1).

Theorem 1.1. *Let $n, t \geq 1$ be integers. Let π be a probability vector on $[n]$ and let $A = (a_{ij})$ be a stochastic $n \times n$ matrix that is reversible with respect to π. Then for all $x_1, \dots, x_n \in \ell_1$ there exist $y_1, \dots, y_n \in \ell_1$ such that*

$$\sum_i \pi_i \|x_i - y_i\|_1^2 + t \sum_{i,j} \pi_i a_{ij} \|y_i - y_j\|_1^2 \leq 3024 \sum_{i,j} \pi_i \Big(\frac{1}{t} \sum_{s=1}^{t} A^s\Big)_{ij} \|x_i - x_j\|_1^2.$$

This is exactly the source statement, with ℓ_1 indexed by $\mathbb{N}_0$.

2 Lemmas

2.1 Markov chain preliminaries

Lemma 2.1 (Powers of stochastic matrices)**.** *If A is a stochastic $n \times n$ matrix, then A^s is stochastic for every $s \in \mathbb{N}_0$.*

Proof. We argue by induction on s. The matrix $A^0 = I$ is stochastic. Suppose A^s is stochastic. Then $(A^{s+1})_{ij} = \sum_k (A^s)_{ik} a_{kj} \geq 0$. Moreover,

$$\sum_j (A^{s+1})_{ij} = \sum_k (A^s)_{ik} \sum_j a_{kj} = \sum_k (A^s)_{ik} = 1. \qquad \square$$

Lemma 2.2 (Weighted Minkowski inequality)**.** *Let ι be a finite set and let $c, f, g, F : \iota \to \mathbb{R}$ be nonnegative with $F \leq f + g$ pointwise. Then*

$$\Big(\sum_i c_i F_i^2\Big)^{1/2} \leq \Big(\sum_i c_i f_i^2\Big)^{1/2} + \Big(\sum_i c_i g_i^2\Big)^{1/2}.$$

Proof. Apply Cauchy–Schwarz to the vectors $(\sqrt{c_i} f_i)$ and $(\sqrt{c_i} g_i)$. This gives

$$\sum_i c_i f_i g_i \le \Big(\sum c_i f_i^2\Big)^{1/2} \Big(\sum c_i g_i^2\Big)^{1/2}.$$

Since $0 \le F_i \le f_i + g_i$, we have $F_i^2 \le (f_i + g_i)^2$. Therefore

$$\sum c_i F_i^2 \le \sum c_i f_i^2 + \sum c_i g_i^2 + 2\sum c_i f_i g_i \le \Big(\big(\sum c_i f_i^2\big)^{1/2} + \big(\sum c_i g_i^2\big)^{1/2}\Big)^2.$$

Taking square roots of these nonnegative quantities gives the claim. □

Lemma 2.3 (Two-step expansion)**.** *Let $\pi \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n\times n}$ be arbitrary, let $f : [n] \times [n] \to \mathbb{R}$, and let $r, s \in \mathbb{N}_0$. Then*

$$\sum_{i,j} \pi_i (A^{r+s})_{ij} f(i,j) = \sum_{(i,k,j)\in[n]^3} \pi_i (A^r)_{ik} (A^s)_{kj}\, f(i,j).$$

Proof. Since $A^{r+s} = A^r A^s$, we have $(A^{r+s})_{ij} = \sum_k (A^r)_{ik}(A^s)_{kj}$. Substitute this, distribute, and rewrite the iterated finite sum as a sum over triples. □

Lemma 2.4 (Marginalizing the last step)**.** *Let A be stochastic, let $\pi \in \mathbb{R}^n$ and $f : [n] \times [n] \to \mathbb{R}$ be arbitrary, and let $r, s \in \mathbb{N}_0$. Then*

$$\sum_{(i,k,j)} \pi_i (A^r)_{ik} (A^s)_{kj} f(i,k) = \sum_{i,k} \pi_i (A^r)_{ik} f(i,k).$$

Proof. Sum over j first. By Lemma 2.1, $\sum_j (A^s)_{kj} = 1$. □

Lemma 2.5 (Marginalizing the first step)**.** *Let A be stochastic and reversible with respect to an arbitrary $\pi \in \mathbb{R}^n$. Let $f : [n] \times [n] \to \mathbb{R}$ and $r, s \in \mathbb{N}_0$. Then*

$$\sum_{(i,k,j)} \pi_i (A^r)_{ik} (A^s)_{kj} f(k,j) = \sum_{k,j} \pi_k (A^s)_{kj} f(k,j).$$

Proof. Reorder the sum with k outermost. The left side becomes

$$\sum_k \Big(\sum_i \pi_i (A^r)_{ik}\Big) \sum_j (A^s)_{kj} f(k,j).$$

By (1), the inner sum over i equals π_k. □

Lemma 2.6 (A geometric second moment)**.** *If $0 \le q < 1$, then the series $\sum_{s\ge 0} s^2 q^s$ converges and its sum is $q(1+q)/(1-q)^3$.*

Proof. The series converges, being a polynomial times a geometric sequence. Let T be its sum. Recall that $\sum_s q^s = (1-q)^{-1}$ and $\sum_s s q^s = q(1-q)^{-2}$. The term with $s = 0$ vanishes, so

$$T = \sum_{s\ge 0} (s+1)^2 q^{s+1} = q \sum_{s\ge 0} (s^2 + 2s + 1) q^s = q\Big(T + \frac{2q}{(1-q)^2} + \frac{1}{1-q}\Big).$$

Hence

$$(1-q)T = q\,\frac{2q + (1-q)}{(1-q)^2} = \frac{q(1+q)}{(1-q)^2}.$$

□

Lemma 2.7 (Square-root subadditivity of the s-step energy)**.** *Assume the following.*

- $\pi \in \mathbb{R}^n$ *has nonnegative entries.*
- A *is stochastic and reversible with respect to* π*.*
- $d : [n] \times [n] \to \mathbb{R}$ *is nonnegative and satisfies* $d(i,k) \le d(i,j) + d(j,k)$ *for all* i, j, k*.*

Define $D(s) = \sum_{i,j} \pi_i (A^s)_{ij}\, d(i,j)^2$ *for* $s \in \mathbb{N}_0$*. Then for all* $r, s \in \mathbb{N}_0$*,*

$$\sqrt{D(r+s)} \le \sqrt{D(r)} + \sqrt{D(s)}.$$

Proof. Index by triples $p = (i,k,j)$ and use the weights $c_p = \pi_i (A^r)_{ik} (A^s)_{kj}$, which are nonnegative by Lemma 2.1. The three energies are sums against the same weights:

- $D(r+s) = \sum_p c_p\, d(i,j)^2$ by Lemma 2.3;
- $D(r) = \sum_p c_p\, d(i,k)^2$ by Lemma 2.4;
- $D(s) = \sum_p c_p\, d(k,j)^2$ by Lemma 2.5.

Apply Lemma 2.2 with $F_p = d(i,j)$, $f_p = d(i,k)$ and $g_p = d(k,j)$. The hypothesis $F \le f + g$ is the triangle inequality. □

2.2 Comparison of geometric and Cesàro averages

Lemma 2.8 (A power estimate)**.** *For every integer* $t \ge 1$*,* $\big(t/(t+1)\big)^t \le 1/2$*.*

Proof. Bernoulli's inequality gives $(1 + 1/t)^t \ge 1 + t \cdot \frac{1}{t} = 2$. Since $t/(t+1) = (1+1/t)^{-1}$, it follows that $\big(t/(t+1)\big)^t \le 1/2$. □

Lemma 2.9 (Geometric sums of periodic sequences)**.** *Let* $t \ge 1$ *be an integer, let* $0 \le q < 1$*, and let* $f : \mathbb{N}_0 \to \mathbb{R}$ *be arbitrary. Then* $\sum_{s \ge 0} q^s f(s \bmod t)$ *converges absolutely, and*

$$\sum_{s \ge 0} q^s f(s \bmod t) = \frac{\sum_{r=0}^{t-1} q^r f(r)}{1 - q^t}.$$

Proof. Let $g(s) = q^s f(s \bmod t)$. Then $|g(s)| \le \big(\sum_{r<t} |f(r)|\big) q^s$, so the series converges absolutely; let S be its sum.

Split off the first t terms. For $s < t$ we have $g(s) = q^s f(s)$. Since $(s+t) \bmod t = s \bmod t$, we have $g(s+t) = q^t g(s)$. Therefore

$$S = \sum_{r<t} q^r f(r) + q^t S.$$

As $q^t < 1$, we may solve for S. □

Lemma 2.10 (Block decomposition)**.** *Let* $t \in \mathbb{N}_0$*. Let* $D : \mathbb{N}_0 \to \mathbb{R}$ *be nonnegative with* $\sqrt{D(r+s)} \le \sqrt{D(r)} + \sqrt{D(s)}$ *for all* r, s*. Then for all* $j, r \in \mathbb{N}_0$*,*

$$D(jt + r) \le 2j^2 D(t) + 2D(r).$$

Proof. By induction on j, $\sqrt{D(jt+r)} \le j\sqrt{D(t)} + \sqrt{D(r)}$. The inductive step is

$$\sqrt{D((jt+r)+t)} \le \sqrt{D(jt+r)} + \sqrt{D(t)}.$$

Both sides of the bound are nonnegative, so we may square it. Then use $(a+b)^2 \le 2a^2 + 2b^2$. □

Lemma 2.11 (The t-step energy against its average)**.** *Let $t \geq 1$. Let $D : \mathbb{N}_0 \to \mathbb{R}$ be nonnegative with $D(0) = 0$ and $\sqrt{D(r+s)} \leq \sqrt{D(r)} + \sqrt{D(s)}$ for all r, s. Then*

$$D(t) \leq 4 \cdot \frac{1}{t} \sum_{s=1}^{t} D(s).$$

Proof. Squaring the hypothesis and using $2\sqrt{D(a)D(b)} \leq D(a)+D(b)$ gives $D(a+b) \leq 2D(a)+2D(b)$. For $0 \leq r \leq t-1$, apply this with $a = r+1$ and $b = t-1-r$:

$$D(t) \leq 2D(r+1) + 2D(t-1-r).$$

Summing over r and reflecting the index in the second sum gives

$$tD(t) \leq 2\sum_{r=0}^{t-1} D(r+1) + 2\sum_{r=0}^{t-1} D(r).$$

The first sum is $\sum_{s=1}^{t} D(s)$. Because $D(0) = 0$ and $D(t) \geq 0$, the second sum is

$$\sum_{r=0}^{t-1} D(r) = \sum_{s=1}^{t-1} D(s) \leq \sum_{s=1}^{t} D(s).$$

Hence $tD(t) \leq 4\sum_{s=1}^{t} D(s)$; divide by t. □

Lemma 2.12 (Second moment of the block index)**.** *Let $t \geq 1$ and $q = t/(t+1)$. Then $\sum_{s\geq 0} q^s \lfloor s/t \rfloor^2$ converges, and*

$$(1-q)\sum_{s\geq 0} q^s \lfloor s/t \rfloor^2 \leq 3.$$

Proof. Since $0 \leq \lfloor s/t \rfloor \leq s/t$, we have $q^s \lfloor s/t \rfloor^2 \leq t^{-2} s^2 q^s$. The comparison series converges by Lemma 2.6, so the left series converges. Again by Lemma 2.6,

$$(1-q)\sum_{s} q^s \lfloor s/t \rfloor^2 \leq \frac{1-q}{t^2} \cdot \frac{q(1+q)}{(1-q)^3}.$$

With $1-q = 1/(t+1)$, $q = t/(t+1)$ and $1+q = (2t+1)/(t+1)$, the right side equals

$$\frac{1}{t^2} \cdot \frac{t(2t+1)}{(t+1)^2} (t+1)^2 = \frac{2t+1}{t} \leq 3.$$ □

Lemma 2.13 (Distribution of the remainder)**.** *Let $t \geq 1$, $q = t/(t+1)$ and $f : \mathbb{N}_0 \to \mathbb{R}$ nonnegative. Then $\sum_{s\geq 0} q^s f(s \bmod t)$ converges, and*

$$(1-q)\sum_{s\geq 0} q^s f(s \bmod t) \leq \frac{2}{t} \sum_{r=0}^{t-1} f(r).$$

Proof. Put $P = \sum_{r<t} q^r f(r)$ and $F = \sum_{r<t} f(r)$, so that $0 \leq P \leq F$. By Lemma 2.9, the series converges with sum $P/(1-q^t)$. By Lemma 2.8, $1-q^t \geq 1/2$, so $P/(1-q^t) \leq 2P \leq 2F$. Finally $(1-q)t = q \leq 1$, which gives

$$(1-q) \cdot 2F \leq \frac{2F}{t}.$$ □

Lemma 2.14 (Geometric versus Cesàro average)**.** *Let $t \geq 1$ and $q = t/(t+1)$. Let $D : \mathbb{N}_0 \to \mathbb{R}$ be nonnegative with $D(0) = 0$ and $\sqrt{D(r+s)} \leq \sqrt{D(r)} + \sqrt{D(s)}$ for all r, s. Then $\sum_{s\geq 0} q^s D(s)$ converges, and*

$$(1-q)\sum_{s\geq 0} q^s D(s) \leq 28 \cdot \frac{1}{t}\sum_{s=1}^{t} D(s).$$

Proof. Let $W = \frac{1}{t}\sum_{s=1}^{t} D(s)$.

Termwise bound. Write $s = \lfloor s/t \rfloor t + (s \bmod t)$. By Lemma 2.10,

$$q^s D(s) \leq 2D(t)\, q^s \lfloor s/t \rfloor^2 + 2\, q^s D(s \bmod t).$$

The right side has a convergent sum by Lemmas 2.12 and 2.13. The terms $q^s D(s)$ are nonnegative, so $\sum_s q^s D(s)$ converges.

Summing. Summing the termwise bound and multiplying by $1 - q \geq 0$ gives

$$(1-q)\sum_{s} q^s D(s) \leq 2D(t) \cdot (1-q)\sum_{s} q^s \lfloor s/t \rfloor^2 + 2(1-q)\sum_{s} q^s D(s \bmod t).$$

We bound the three ingredients:

- $D(t) \leq 4W$ by Lemma 2.11;
- the first series term is at most 3 by Lemma 2.12;
- the second series term is at most $\frac{2}{t}\sum_{r<t} D(r)$ by Lemma 2.13, and $\frac{2}{t}\sum_{r<t} D(r) \leq 2W$ because $\sum_{r<t} D(r) \leq \sum_{s=1}^{t} D(s)$ (as $D(0) = 0$ and $D(t) \geq 0$).

Altogether the total is at most $2 \cdot 4W \cdot 3 + 2 \cdot 2W = 28W$. □

2.3 The resolvent

In the next lemmas, for a stochastic A and $0 \leq q < 1$, G denotes the matrix

$$G_{ij} = \sum_{s\geq 0} (1-q) q^s (A^s)_{ij}.$$

The series converges by Lemma 2.15.

Lemma 2.15 (Convergence)**.** *If A is stochastic and $0 \leq q < 1$, then for all i, j the series $\sum_{s\geq 0}(1-q)q^s(A^s)_{ij}$ converges.*

Proof. By Lemma 2.1, every entry of A^s lies in $[0, 1]$: it is nonnegative and at most its row sum 1. So the terms lie between 0 and $(1-q)q^s$, which is summable. □

Lemma 2.16 (Resolvent comparison)**.** *Let $t \geq 1$. Let π be a probability vector on $[n]$ and let A be stochastic and reversible with respect to π. Let $x_1, \ldots, x_n \in \ell_1$, let $q = t/(t+1)$, and let G be as above. Then*

$$\sum_{i,j} \pi_i G_{ij} \|x_i - x_j\|_1^2 \leq 28 \sum_{i,j} \pi_i \Big(\frac{1}{t}\sum_{s=1}^{t} (A^s)_{ij}\Big) \|x_i - x_j\|_1^2.$$

Proof. Let $D(s) = \sum_{i,j} \pi_i (A^s)_{ij} \|x_i - x_j\|_1^2$. We check the hypotheses of Lemma 2.14:

- $D(0) = \sum_i \pi_i \|x_i - x_i\|_1^2 = 0$;

- $D \geq 0$ because $\pi_i \geq 0$ and $(A^s)_{ij} \geq 0$ (Lemma 2.1);
- D is square-root subadditive by Lemma 2.7 with $d(i,j) = \|x_i - x_j\|_1$, which is nonnegative and satisfies the triangle inequality.

Left side. For fixed i, j, the series $\sum_s \pi_i(1-q)q^s(A^s)_{ij}\|x_i - x_j\|_1^2$ converges to $\pi_i G_{ij}\|x_i - x_j\|_1^2$. Summing these finitely many convergent series shows that $\sum_s (1-q)q^s D(s)$ converges to the left side. Hence the left side equals $(1-q)\sum_s q^s D(s)$.

Right side. Exchanging finite sums shows that the right side without the factor 28 equals $\frac{1}{t}\sum_{s=1}^{t} D(s)$.

The claim now follows from Lemma 2.14. □

Lemma 2.17 (Nonnegativity)**.** *If A is stochastic and $0 \leq q < 1$, then $G_{ij} \geq 0$ for all i, j.*

Proof. Each term $(1-q)q^s(A^s)_{ij}$ is nonnegative by Lemma 2.1. □

Lemma 2.18 (Row sums)**.** *If A is stochastic and $0 \leq q < 1$, then $\sum_j G_{ij} = 1$ for every i.*

Proof. Exchange the finite sum over j with the convergent series. Using Lemma 2.1,

$$\sum_j G_{ij} = \sum_s (1-q)q^s \sum_j (A^s)_{ij} = \sum_s (1-q)q^s = 1.$$ □

Lemma 2.19 (Stationarity of the resolvent)**.** *Let $\pi \in \mathbb{R}^n$ be arbitrary, let A be stochastic and reversible with respect to π, and let $0 \leq q < 1$. Then $\sum_i \pi_i G_{ij} = \pi_j$ for every j.*

Proof. Exchange the finite sum with the series and use (1):

$$\sum_i \pi_i G_{ij} = \sum_s (1-q)q^s \sum_i \pi_i (A^s)_{ij} = \sum_s (1-q)q^s \pi_j = \pi_j.$$ □

Lemma 2.20 (First-step identity)**.** *If A is stochastic and $0 \leq q < 1$, then for all i, j,*

$$G_{ij} = (1-q)\delta_{ij} + q\sum_l a_{il} G_{lj}.$$

Proof. Split off the term $s = 0$, which is $(1-q)\delta_{ij}$. In the remaining terms use $(A^{s+1})_{ij} = \sum_l a_{il}(A^s)_{lj}$:

$$\sum_{s\geq 0} (1-q)q^{s+1}(A^{s+1})_{ij} = q\sum_{s\geq 0}\sum_l a_{il}(1-q)q^s(A^s)_{lj} = q\sum_l a_{il} G_{lj}.$$

The finite sum over l may be exchanged with the convergent series (Lemma 2.15). □

Lemma 2.21 (Last-step identity)**.** *If A is stochastic and $0 \leq q < 1$, then for all i, j,*

$$G_{ij} = (1-q)\delta_{ij} + q\sum_l G_{il} a_{lj}.$$

Proof. The same argument as for Lemma 2.20, now with $(A^{s+1})_{ij} = \sum_l (A^s)_{il} a_{lj}$. The series $\sum_s \sum_l (1-q)q^s (A^s)_{il} a_{lj}$ converges to $\sum_l G_{il} a_{lj}$, so the shifted series sums to $q\sum_l G_{il} a_{lj}$. □

2.4 The flat map

Let $\phi(r) = 3r^2 - 2r^3$ for $r \in \mathbb{R}$. Note that $\phi(0) = 0$ and $\phi(1) = 1$.

Lemma 2.22 (Scalar increment)**.** *Let $r, r' \in [0,1]$ and $d \in \{0,1\}$. Then*

$$|\phi(r') - \phi(r)| \le 6|r-d|\,|r'-r| + 3(r'-r)^2.$$

Proof. Put $\delta = r' - r$. Direct expansion gives the factorization

$$\phi(r') - \phi(r) = \delta(6r(1-r) + \delta(3 - 4r - 2r')).$$

We bound the two terms in the bracket:

- $0 \le 6r(1-r) \le 6|r-d|$, since $r(1-r) \le r$ and $r(1-r) \le 1-r$;
- $|3 - 4r - 2r'| \le 3$, since $4r + 2r' \in [0,6]$.

Hence $|\phi(r') - \phi(r)| \le |\delta|\,(6|r-d| + 3|\delta|)$, which is the claim. □

Lemma 2.23 (Weighted increment)**.** *Let ι be finite and let $w : \iota \to \mathbb{R}$ be nonnegative. Let $e \in \{0,1\}^\iota$ and $u, u' \in [0,1]^\iota$. Set*

$$P = \sum_B w_B(u_B - e_B)^2, \qquad Y = \sum_B w_B(u'_B - u_B)^2.$$

Then

$$\Big(\sum_B w_B|\phi(u'_B) - \phi(u_B)|\Big)^2 \le 72\,PY + 18\,Y^2.$$

Proof. Let S be the sum in parentheses and let $X = \sum_B w_B|u_B - e_B|\,|u'_B - u_B|$. Multiplying Lemma 2.22 by w_B and summing gives $0 \le S \le 6X + 3Y$. Cauchy–Schwarz with the weights w_B gives $X^2 \le PY$. Therefore

$$S^2 \le (6X + 3Y)^2 \le 72X^2 + 18Y^2 \le 72PY + 18Y^2.$$ □

Lemma 2.24 (Flat inequality)**.** *Under the hypotheses of Lemma 2.23, set $P' = \sum_B w_B(u'_B - e_B)^2$ and $L = \sum_B w_B(u_B - e_B)(u'_B - u_B)$. Then*

$$\Big(\sum_B w_B|\phi(u'_B) - \phi(u_B)|\Big)^2 \le 108(P'^2 - P^2 - 4PL).$$

Proof. We have $P' = P + 2L + Y$, so

$$R := P'^2 - P^2 - 4PL = 2PY + (2L + Y)^2.$$

By Cauchy–Schwarz, $L^2 \le PY$. Hence

$$Y^2 = \big((2L + Y) - 2L\big)^2 \le 2(2L + Y)^2 + 8L^2 \le 2(2L + Y)^2 + 8PY.$$

Using Lemma 2.23, the left side is at most

$$72PY + 18Y^2 \le 216PY + 36(2L + Y)^2 \le 108R.$$ □

Lemma 2.25 (One-step balance)**.** *Assume the following.*

- ι *is a finite set,* $m \in \mathbb{N}_0$*, and* $w : \iota \to \mathbb{R}$ *is nonnegative.*
- $e, u, v \in \mathbb{R}^\iota$ *and* $U_1, \dots, U_m \in \mathbb{R}^\iota$*.*
- $p \geq 0$*,* $c_1, \dots, c_m \geq 0$*, and* $p + \sum_j c_j = 1$*.*
- $u = pv + \sum_j c_j U_j$ *coordinatewise.*
- $e \in \{0,1\}^\iota$ *and* $u, v, U_j \in [0,1]^\iota$*.*

For $a \in \mathbb{R}^\iota$ *write* $\Psi(a) = \big(\sum_B w_B (a_B - e_B)^2\big)^2$*. Then*

$$p\Big(\sum_B w_B |\phi(v_B) - \phi(u_B)|\Big)^2 + \sum_j c_j \Big(\sum_B w_B |\phi(U_{j,B}) - \phi(u_B)|\Big)^2 \leq 108\Big(p\Psi(v) + \sum_j c_j \Psi(U_j) - \Psi(u)\Big).$$

Proof. Let $S_0 = \sum_B w_B (u_B - e_B)^2$, so that $\Psi(u) = S_0^2$. Let L_v and L_j be the quantity L of Lemma 2.24 for the pairs (u, v) and (u, U_j).

The linear terms cancel. Coordinatewise,

$$p(v_B - u_B) + \sum_j c_j (U_{j,B} - u_B) = pv_B + \sum_j c_j U_{j,B} - \Big(p + \sum_j c_j\Big) u_B = 0.$$

Multiplying by $w_B(u_B - e_B)$ and summing over B gives $pL_v + \sum_j c_j L_j = 0$.

Combining. Apply Lemma 2.24 to (u, v) and to each (u, U_j). Multiply by p and by c_j respectively, and add. The left side is bounded by

$$108\Big(p\Psi(v) + \sum_j c_j \Psi(U_j) - \Big(p + \sum_j c_j\Big) S_0^2 - 4S_0\Big(pL_v + \sum_j c_j L_j\Big)\Big).$$

Since $p + \sum_j c_j = 1$ and $pL_v + \sum_j c_j L_j = 0$, this is the claim. □

Lemma 2.26 (Summation against the resolvent)**.** *Assume the following.*

- $\pi \in \mathbb{R}^n$ *is nonnegative,* $A, G \in \mathbb{R}^{n \times n}$*, and* $p, q \in \mathbb{R}$ *with* $p \geq 0$*.*
- $G \geq 0$ *entrywise and* $\sum_i \pi_i G_{ij} = \pi_j$ *for all* j*.*
- $G_{kj} = p\delta_{kj} + q \sum_i G_{ki} a_{ij}$ *for all* k, j*.*
- $J_d : [n] \to \mathbb{R}$ *and* $J_h, F_z, F_h : [n] \times [n] \to \mathbb{R}$*, with* $F_h \geq 0$*.*
- *For all* k, i*,*

$$pJ_d(i) + q \sum_j a_{ij} J_h(i,j) \leq 108\Big(pF_z(k,i) + q \sum_j a_{ij} F_h(k,j) - F_h(k,i)\Big).$$

Then

$$\sum_i \pi_i \Big(pJ_d(i) + q \sum_j a_{ij} J_h(i,j)\Big) \leq 108\, p \sum_{k,i} \pi_k G_{ki} F_z(k,i).$$

Proof. Write $J(i) = pJ_d(i) + q\sum_j a_{ij}J_h(i,j)$ and fix k.

Telescoping identity. By the identity for G,

$$\sum_i G_{ki}\, q\sum_j a_{ij}F_h(k,j) = \sum_j \Big(q\sum_i G_{ki}a_{ij}\Big)F_h(k,j) = \sum_j G_{kj}F_h(k,j) - pF_h(k,k).$$

Bound for fixed k. Multiply the hypothesis by $\pi_k G_{ki} \geq 0$ and sum over i. By the telescoping identity, the terms involving F_h reduce to $-pF_h(k,k) \leq 0$, giving

$$\sum_i \pi_k G_{ki}J(i) \leq 108\,\pi_k\Big(p\sum_i G_{ki}F_z(k,i) - pF_h(k,k)\Big) \leq 108\,p\sum_i \pi_k G_{ki}F_z(k,i).$$

Summing over k. Since $\pi_i = \sum_k \pi_k G_{ki}$,

$$\sum_i \pi_i J(i) = \sum_k\sum_i \pi_k G_{ki}J(i),$$

and summing the previous bound over k gives the claim. □

Lemma 2.27 (The harmonic extension)**.** *Assume the following.*

- *ι is finite, $G, A \in \mathbb{R}^{n\times n}$, and $p, q \in \mathbb{R}$.*
- *$G \geq 0$, $\sum_j G_{ij} = 1$ for all i, and $G_{ij} = p\delta_{ij} + q\sum_l a_{il}G_{lj}$ for all i, j.*
- *$z_1, \ldots, z_n \in [0,1]^\iota$.*

Define $h_i \in \mathbb{R}^\iota$ by $h_i(B) = \sum_j G_{ij}z_j(B)$. Then $h_i \in [0,1]^\iota$ and

$$h_i(B) = pz_i(B) + \sum_j (qa_{ij})h_j(B) \qquad \text{for all } i, B.$$

Proof. Each $h_i(B)$ is a convex combination of numbers in $[0,1]$, because $G \geq 0$ and its rows sum to 1. For the identity, substitute the relation for G_{ik}:

$$h_i(B) = \sum_k \Big(p\delta_{ik} + q\sum_l a_{il}G_{lk}\Big)z_k(B) = pz_i(B) + \sum_l qa_{il}\sum_k G_{lk}z_k(B) = pz_i(B) + \sum_l qa_{il}h_l(B).$$ □

2.5 Cut decompositions

For $a : [n] \to \mathbb{R}$ we say that $b : \mathcal{P}_n \to \mathbb{R}$ is an *upper cut decomposition of a* if the following three conditions hold.

1. $b(B) \geq 0$ for all B.
2. Whenever $b(B) \neq 0$: for all i, j, if $j \in B$ and $a_j \leq a_i$, then $i \in B$.
3. $a_i - a_j = \sum_{B\in\mathcal{P}_n} b(B)\big(\mathbf{1}_B(i) - \mathbf{1}_B(j)\big)$ for all i, j.

Lemma 2.28 (Adding the lowest cut)**.** *Let $a : [n] \to \mathbb{R}$ and let $m < c$ be reals with $m \leq a_i$ for all i, and $a_i = m$ whenever $a_i < c$. Suppose b' is an upper cut decomposition of $a' = (\max(a_i, c))_i$. Then a has an upper cut decomposition.*

Proof. Let $B^* = \{i : c \le a_i\}$. For every k,

$$a_k = \max(a_k, c) + (c - m)\big(\mathbf{1}_{B^*}(k) - 1\big).$$

Indeed, if $c \le a_k$ both sides equal a_k; otherwise $a_k = m$ and the right side is $c - (c - m)$.

Set $b(B) = b'(B) + (c - m)\mathbf{1}_{\{B = B^*\}}$. We check the three conditions.

1. $b \ge 0$, since $b' \ge 0$ and $c > m$.
2. Suppose $b(B) \ne 0$, $j \in B$ and $a_j \le a_i$. If $B = B^*$, then $c \le a_j \le a_i$, so $i \in B^*$. Otherwise $b'(B) = b(B) \ne 0$; moreover $a_j \le a_i$ implies $\max(a_j, c) \le \max(a_i, c)$, so $i \in B$ by the property of b'.
3. By the property of b' and the formula for a_k,

$$\sum_B b(B)\big(\mathbf{1}_B(i) - \mathbf{1}_B(j)\big) = \max(a_i, c) - \max(a_j, c) + (c - m)\big(\mathbf{1}_{B^*}(i) - \mathbf{1}_{B^*}(j)\big) = a_i - a_j. \qquad \square$$

Lemma 2.29 (Merging the two lowest values)**.** *Let $a : [n] \to \mathbb{R}$ with $a_i \ne a_j$ for some i, j. Then there are reals $m < c$ with the following properties:*

- *$m \le a_i$ for all i;*
- *$a_i < c$ implies $a_i = m$;*
- *the set $\{\max(a_i, c) : i \in [n]\}$ has strictly fewer elements than $\{a_i : i \in [n]\}$.*

Proof. Let $V = \{a_i\}$ and $m = \min V$. Since a is not constant, $V \setminus \{m\} \ne \emptyset$; let $c = \min(V \setminus \{m\}) > m$. If $a_i < c$, then $a_i = m$, for otherwise $a_i \in V \setminus \{m\}$ and $c \le a_i$.

We claim $\{\max(a_i, c)\} = V \setminus \{m\}$, which gives the cardinality statement.

- If $a_i \le c$, then $\max(a_i, c) = c \in V \setminus \{m\}$. Otherwise $\max(a_i, c) = a_i > m$.
- Conversely, let $a_i \in V \setminus \{m\}$. If $a_i \le c$, then $a_i = c$, since $a_i < c$ would force $a_i = m$. In either case $\max(a_i, c) = a_i$. $\square$

Lemma 2.30 (Existence of upper cut decompositions)**.** *Every $a : [n] \to \mathbb{R}$ has an upper cut decomposition.*

Proof. We prove, by induction on $k \in \mathbb{N}_0$, that every a taking at most k distinct values has one.

If a is constant, $b = 0$ works: condition 2 is vacuous and condition 3 reads $0 = 0$. This covers $n = 0$.

Now let a be non-constant and take m, c from Lemma 2.29. For $k = 0$ this is impossible, as $a' = \max(a, c)$ would take fewer than 0 values. For $k + 1$, the function a' takes at most k values, so it has an upper cut decomposition by the induction hypothesis. Lemma 2.28 then yields one for a. $\square$

Lemma 2.31 (Signed and absolute cut identities)**.** *For every $a : [n] \to \mathbb{R}$ there is a nonnegative $b : \mathcal{P}_n \to \mathbb{R}$ such that for all i, j,*

$$a_i - a_j = \sum_B b(B)\big(\mathbf{1}_B(i) - \mathbf{1}_B(j)\big), \qquad |a_i - a_j| = \sum_B b(B)\big|\mathbf{1}_B(i) - \mathbf{1}_B(j)\big|.$$

Proof. Take b from Lemma 2.30. Suppose $a_j \le a_i$. We claim every term $b(B)(\mathbf{1}_B(i) - \mathbf{1}_B(j))$ equals $b(B)|\mathbf{1}_B(i) - \mathbf{1}_B(j)|$:

- this is trivial if $b(B) = 0$;
- if $b(B) \neq 0$ and $j \in B$, then $i \in B$, so the difference is 0;
- if $j \notin B$, the difference is $\mathbf{1}_B(i) \geq 0$.

So $|a_i - a_j| = a_i - a_j$ gives the absolute identity. The case $a_i \leq a_j$ follows by symmetry. □

Lemma 2.32 (Bounded cut weights)**.** *For every $a : [n] \to \mathbb{R}$ there is $b : \mathcal{P}_n \to \mathbb{R}$ with $0 \leq b(B) \leq 2\sum_i |a_i|$ for all B, such that both identities of Lemma 2.31 hold.*

Proof. Take b from Lemma 2.31. Let $\tilde{b}(B) = b(B)$ if B is nonempty and $B \neq [n]$, and $\tilde{b}(B) = 0$ otherwise. For $B = \emptyset$ or $B = [n]$, the factor $\mathbf{1}_B(i) - \mathbf{1}_B(j)$ vanishes, so both identities persist for $\tilde{b}$.

For a nontrivial B, pick $i \in B$ and $j \notin B$. Keeping only the term B in the absolute identity gives

$$\tilde{b}(B) \leq |a_i - a_j| \leq |a_i| + |a_j| \leq 2\sum_k |a_k|.$$ □

Lemma 2.33 (Extensionality in ℓ_1)**.** *If $f, g \in \ell_1$ satisfy $f(k) = g(k)$ for all k, then $f = g$.*

Proof. Elements of ℓ_1 are sequences. □

Lemma 2.34 (Coordinates of finite combinations)**.** *Let ι be finite, $x_0 \in \ell_1$, $c : \iota \to \mathbb{R}$ and $\beta : \iota \to \ell_1$. Then for every k,*

$$\Big(x_0 + \sum_B c_B \beta_B\Big)(k) = x_0(k) + \sum_B c_B \beta_B(k).$$

Proof. The vector space operations of ℓ_1 are coordinatewise. □

2.6 Assembly in the cut space

Lemma 2.35 (Real assembly)**.** *Assume the following.*

- *$\pi \in \mathbb{R}^n$ is nonnegative and A is stochastic.*
- *ι is finite, $w : \iota \to \mathbb{R}$ is nonnegative, and $z_1, \ldots, z_n \in \{0,1\}^\iota$.*
- *$0 \leq q < 1$ and $\tau(1-q) = q$.*
- *$G \in \mathbb{R}^{n \times n}$ satisfies: $G \geq 0$; $\sum_j G_{ij} = 1$ for all i; $\sum_i \pi_i G_{ij} = \pi_j$ for all j; and the identities of Lemmas 2.20 and 2.21.*

Then there are $h_1, \ldots, h_n \in \mathbb{R}^\iota$ such that

$$\sum_i \pi_i \Big(\sum_B w_B |z_i(B) - \phi(h_i(B))|\Big)^2 + \tau \sum_{i,j} \pi_i a_{ij} \Big(\sum_B w_B |\phi(h_i(B)) - \phi(h_j(B))|\Big)^2 \leq 108 \sum_{i,j} \pi_i G_{ij} \Big(\sum_B w_B |z_i(B) - z_j(B)|\Big)^2.$$

Proof. Let $h_i(B) = \sum_j G_{ij} z_j(B)$. By Lemma 2.27 with $p = 1 - q$, we have $h_i \in [0,1]^\iota$ and

$$h_i = (1-q) z_i + \sum_j q a_{ij} h_j.$$

Local inequality. Fix k, i and apply Lemma 2.25 with

$$e = z_k, \quad u = h_i, \quad v = z_i, \quad U_j = h_j \ (m = n), \quad p = 1 - q, \quad c_j = qa_{ij}.$$

The hypotheses hold: $1 - q \geq 0$, $c_j \geq 0$, $(1 - q) + \sum_j qa_{ij} = 1$, the balance relation was just shown, and all vectors lie in $[0, 1]^\iota$. For binary coordinates we have $\phi(z_i(B)) = z_i(B)$ and $(z_i(B) - z_k(B))^2 = |z_k(B) - z_i(B)|$. Set

$$J_d(i) = \Big(\sum_B w_B |z_i(B) - \phi(h_i(B))|\Big)^2, \qquad J_h(i,j) = \Big(\sum_B w_B |\phi(h_i(B)) - \phi(h_j(B))|\Big)^2,$$
$$F_z(k,i) = \Big(\sum_B w_B |z_k(B) - z_i(B)|\Big)^2, \qquad F_h(k,i) = \Big(\sum_B w_B (h_i(B) - z_k(B))^2\Big)^2 \geq 0.$$

With this notation, Lemma 2.25 reads

$$(1 - q)J_d(i) + q\sum_j a_{ij} J_h(i,j) \leq 108\Big((1 - q)F_z(k,i) + q\sum_j a_{ij} F_h(k,j) - F_h(k,i)\Big).$$

Summation. Lemma 2.26 with $p = 1 - q$ uses the last-step identity, $G \geq 0$ and $\pi G = \pi$. Writing $S_1 = \sum_i \pi_i J_d(i)$, $S_2 = \sum_{i,j} \pi_i a_{ij} J_h(i,j)$ and $R = \sum_{k,i} \pi_k G_{ki} F_z(k,i)$, it gives

$$(1 - q)S_1 + qS_2 \leq 108(1 - q)R.$$

Since $q = \tau(1 - q)$, the left side equals $(1 - q)(S_1 + \tau S_2)$. Dividing by $1 - q > 0$ gives $S_1 + \tau S_2 \leq 108R$, which is the claim after renaming indices in R. □

2.7 The ℓ_1 cut representation

Lemma 2.36 (Norm of a cut combination)**.** *Let ι be finite. Let $f \in \ell_1$, let $\beta : \iota \to \ell_1$ with $\beta_B(k) \geq 0$ for all B, k, and let $d : \iota \to \mathbb{R}$. If $|f(k)| = \sum_B \beta_B(k) d_B$ for every k, then $\|f\|_1 = \sum_B \|\beta_B\|_1 d_B$.*

Proof. The series $\sum_k \beta_B(k)$ converge and $\|\beta_B\|_1 = \sum_k \beta_B(k)$, since $\beta_B \geq 0$. Exchanging the finite sum over B with the series over k,

$$\|f\|_1 = \sum_k \sum_B \beta_B(k) d_B = \sum_B d_B \sum_k \beta_B(k) = \sum_B \|\beta_B\|_1 d_B.$$ □

Lemma 2.37 (Cut vectors)**.** *For $x_1, \ldots, x_n \in \ell_1$ there are $\beta_B \in \ell_1$ $(B \in \mathcal{P}_n)$ with $\beta_B(k) \geq 0$, such that for all i, j, k:*

$$x_i(k) = x_j(k) + \sum_B (\mathbf{1}_B(i) - \mathbf{1}_B(j))\beta_B(k),$$
$$|(x_i - x_j)(k)| = \sum_B \beta_B(k) |\mathbf{1}_B(i) - \mathbf{1}_B(j)|.$$

Proof. For each coordinate k, choose $b_k : \mathcal{P}_n \to \mathbb{R}$ by Lemma 2.32 applied to $a = (x_i(k))_i$, and set $\beta_B(k) = b_k(B)$. Then

$$0 \leq \beta_B(k) \leq 2\sum_i |x_i(k)|.$$

The right side is summable in k, since each $x_i \in \ell_1$, so $\beta_B \in \ell_1$. The two identities are the two identities of Lemma 2.32 at coordinate k. □

Lemma 2.38 (Cut representation with ambient reconstruction)**.** *Let $n \geq 1$ and $x_1, \dots, x_n \in \ell_1$. There exist*

- *weights $w : \mathcal{P}_n \to \mathbb{R}$ with $w \geq 0$,*
- *binary vectors $z_1, \dots, z_n \in \{0,1\}^{\mathcal{P}_n}$, and*
- *a map $T : \mathbb{R}^{\mathcal{P}_n} \to \ell_1$*

such that for all i, j and all $u, u' \in \mathbb{R}^{\mathcal{P}_n}$:

- $T(z_i) = x_i$;
- $\|T(u) - T(u')\|_1 \leq \sum_B w_B |u_B - u'_B|$;
- $\|x_i - x_j\|_1 = \sum_B w_B |z_i(B) - z_j(B)|$.

Proof. Take β_B from Lemma 2.37 and fix the index $i_0 = 1$. Set

$$w_B = \|\beta_B\|_1, \qquad z_i(B) = \mathbf{1}_B(i), \qquad T(u) = x_{i_0} + \sum_B (u_B - \mathbf{1}_B(i_0))\beta_B.$$

- By Lemma 2.34, $T(z_i)(k) = x_{i_0}(k) + \sum_B (\mathbf{1}_B(i) - \mathbf{1}_B(i_0))\beta_B(k) = x_i(k)$. Hence $T(z_i) = x_i$ by Lemma 2.33.
- $T(u) - T(u') = \sum_B (u_B - u'_B)\beta_B$, and the triangle inequality gives the contraction bound.
- The distance identity is Lemma 2.36 with $f = x_i - x_j$ and $d_B = |\mathbf{1}_B(i) - \mathbf{1}_B(j)|$. □

Lemma 2.39 (Assembly in ℓ_1)**.** *Assume the hypotheses of Lemma 2.35 on $\pi, A, \iota, w, z, q, \tau, G$. Assume further that $x_1, \dots, x_n \in \ell_1$ and $T : \mathbb{R}^\iota \to \ell_1$ satisfy, for all i, j and all $u, u' \in \mathbb{R}^\iota$:*

- $T(z_i) = x_i$;
- $\|T(u) - T(u')\|_1 \leq \sum_B w_B |u_B - u'_B|$;
- $\|x_i - x_j\|_1 = \sum_B w_B |z_i(B) - z_j(B)|$.

Then there are $y_1, \dots, y_n \in \ell_1$ with

$$\sum_i \pi_i \|x_i - y_i\|_1^2 + \tau \sum_{i,j} \pi_i a_{ij} \|y_i - y_j\|_1^2 \leq 108 \sum_{i,j} \pi_i G_{ij} \|x_i - x_j\|_1^2.$$

Proof. Take h_i from Lemma 2.35 and set $y_i = T((\phi(h_i(B)))_B)$.

Sign of τ. We have $\tau \geq 0$: if $\tau < 0$, then $q = \tau(1-q) < 0$, since $1 - q > 0$, a contradiction.

Comparison. By $T(z_i) = x_i$ and the contraction property,

$$\|x_i - y_i\|_1 \leq \sum_B w_B |z_i(B) - \phi(h_i(B))|, \qquad \|y_i - y_j\|_1 \leq \sum_B w_B |\phi(h_i(B)) - \phi(h_j(B))|.$$

Square these nonnegative bounds and multiply by $\pi_i \geq 0$, by $\pi_i a_{ij} \geq 0$ and by $\tau \geq 0$ respectively. This shows that the left side is at most the left side of Lemma 2.35. Finally, the distance identity converts the right side of that lemma into $108 \sum_{i,j} \pi_i G_{ij} \|x_i - x_j\|_1^2$. □

3 Proof of the theorem

Proof of Theorem 1.1. Let $n, t \geq 1$, π, A and $x_1, \dots, x_n$ be as in the theorem.

Step 1: cut representation. Apply Lemma 2.38 to obtain w, z, T with index set $\iota = \mathcal{P}_n$.

Step 2: parameters. Put $q = t/(t+1)$. Then $0 \leq q < 1$ and $t(1-q) = q$, so $\tau = t$ is admissible.

Step 3: the resolvent. Let $G_{ij} = \sum_{s \geq 0}(1-q)q^s(A^s)_{ij}$. By the lemmas on the resolvent:

- $G \geq 0$ (Lemma 2.17);
- every row of G sums to 1 (Lemma 2.18);
- $\pi G = \pi$ (Lemma 2.19);
- G satisfies the first-step and last-step identities (Lemmas 2.20 and 2.21).

Step 4: construction. Lemma 2.39, using only $\pi_i \geq 0$, gives $y_1, \dots, y_n \in \ell_1$ with

$$\sum_i \pi_i \|x_i - y_i\|_1^2 + t \sum_{i,j} \pi_i a_{ij} \|y_i - y_j\|_1^2 \leq 108 \sum_{i,j} \pi_i G_{ij} \|x_i - x_j\|_1^2.$$

Step 5: comparison. By Lemma 2.16, the right side is at most

$$108 \cdot 28 \sum_{i,j} \pi_i \Big(\frac{1}{t} \sum_{s=1}^{t} (A^s)_{ij} \Big) \|x_i - x_j\|_1^2.$$

Since $108 \cdot 28 = 3024$, this completes the proof. □

Relation to the source proof. The construction is the same as in the source proof. The points are $y_i = T(\Phi(h_i))$, where $h_i = \sum_j G_{ij} z_j$ is the resolvent average of the binary cut encodings and Φ applies $\phi(r) = 3r^2 - 2r^3$ coordinatewise. The constants are also the same: 108 from the flat inequality and $28 = 2 \cdot 3 \cdot 4 + 2 \cdot 2$ from the comparison. The formal argument deviates in the following ways.

1. *No probability is used.* The source constructs a killed Markov chain and an H-valued martingale, and applies optional stopping together with bounded and monotone convergence. Here row k of G plays the role of the law of the death position for the chain started at k. The vanishing of the conditional linear term becomes the exact cancellation $pL_v + \sum_j c_j L_j = 0$ in Lemma 2.25, applied with a centre $e = z_k$ depending on the starting state. The telescoping over time becomes the identity $G = (1-q)I + qGA$ in Lemma 2.26. The jump identity "$= B + tE_A$" of the source appears as the factorization $(1-q)S_1 + qS_2 = (1-q)(S_1 + tS_2)$.

2. *Flat inequality.* The scalar estimate is proved by the exact factorization in Lemma 2.22, not by integrating ϕ'.

3. *Comparison.* The comparison is proved through explicit generating-function sums (Lemmas 2.6 and 2.9), not through the random variables $J = \lfloor S/t \rfloor$ and $R = S - tJ$.

4. *Cut representation.* The cut representation is obtained by an induction on the number of distinct values in each coordinate, with a choice of decomposition in each coordinate. It is not given by the explicit formula $b_B(k) = (\min_B x_i(k) - \max_{B^c} x_i(k))_+$. All 2^n subsets are used with nonnegative, possibly zero, weights, so the degenerate case of coinciding points needs no separate treatment.

5. *Hypotheses used.* As in the source remark, reversibility is used only through the stationarity (1). Of the probability-vector hypothesis, only the nonnegativity of the entries of π enters the argument.

A Correspondence with the formal proof

Each lemma of the text and the Lean declaration that states it; Theorem 1.1 is `openaimath_332_metricmarkovcotypel1`

Lemma	Lean declaration	Status
Lemma 2.1	`stoch_pow`	verified
Lemma 2.2	`weighted_minkowski`	verified
Lemma 2.3	`markov_subadd_expand`	verified
Lemma 2.4	`markov_subadd_left`	verified
Lemma 2.5	`markov_subadd_right`	verified
Lemma 2.6	`comp_geom_sq_sqsum`	verified
Lemma 2.7	`markov_subadd`	verified
Lemma 2.8	`comp_geom_mod_pow_le`	verified
Lemma 2.9	`comp_geom_mod_hasSum`	verified
Lemma 2.10	`comp_block`	verified
Lemma 2.11	`comp_Dt`	verified
Lemma 2.12	`comp_geom_sq`	verified
Lemma 2.13	`comp_geom_mod`	verified
Lemma 2.14	`comp_real`	verified
Lemma 2.15	`res_summable`	verified
Lemma 2.16	`res_comparison`	verified
Lemma 2.17	`res_nonneg`	verified
Lemma 2.18	`res_rowsum`	verified
Lemma 2.19	`res_stationary`	verified
Lemma 2.20	`res_left`	verified
Lemma 2.21	`res_right`	verified
Lemma 2.22	`flat_scalar`	verified
Lemma 2.23	`flat_increment`	verified
Lemma 2.24	`flat_ineq`	verified
Lemma 2.25	`local_ineq`	verified
Lemma 2.26	`global_sum`	verified
Lemma 2.27	`h_props`	verified
Lemma 2.28	`cut_coord_upper_shift`	verified
Lemma 2.29	`cut_coord_upper_reduce`	verified
Lemma 2.30	`cut_coord_upper`	verified
Lemma 2.31	`cut_coord_core`	verified
Lemma 2.32	`cut_coord`	verified
Lemma 2.33	`cut_l1_ext`	verified
Lemma 2.34	`cut_l1_apply`	verified
Lemma 2.35	`assembly_real`	verified
Lemma 2.36	`cut_l1_norm_combo`	verified
Lemma 2.37	`cut_l1_vectors`	verified
Lemma 2.38	`cut_l1`	verified
Lemma 2.39	`assembly`	verified